

RS-1 Atlatl lower stage

Technical specifications and assembly instructions

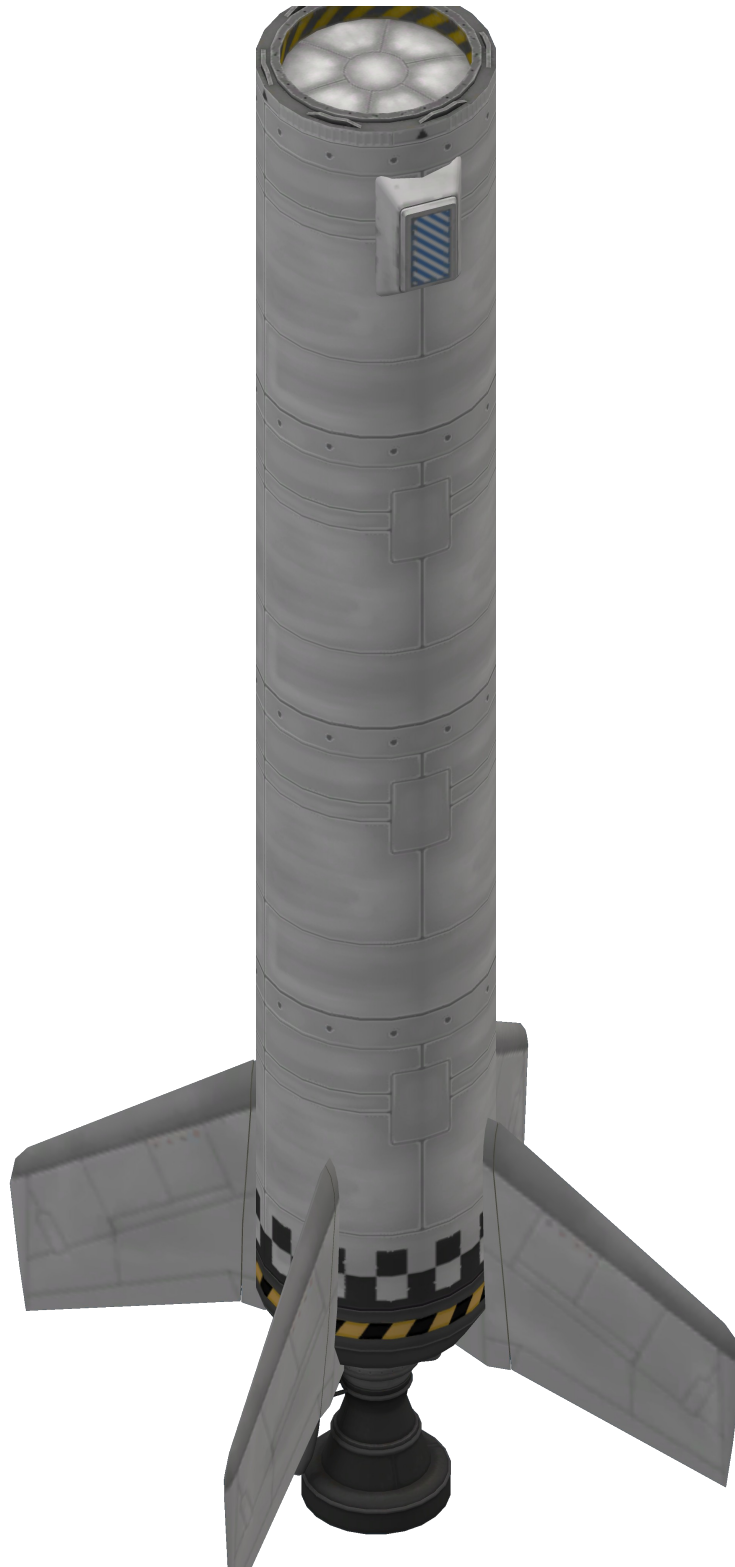


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Description

The RS-1 Atlatl stage is a low tech, Reliant-based dumb launch stage. Optimized for low atmosphere flight. The stage comes in different variants differing in propellant capacity and is fully recoverable using built-in parachutes.

This stage has no onboard computer and control is delegated to either the upper stage computer or the payload. Stage recovery is completely passive.

Basic construction

The stage is constructed using two subassemblies. The lower part of the stage uses a subassembly with the engine, a fuel tank and 4 fins. The fins are set at 45 degrees from yaw and pitch planes. The uppermost part of the stage is created using a parachute subassembly. It contains a fuel tank, a pair of parachutes and a TD-12 decoupler. Between the two subassemblies a number of T400 tanks can be installed. This number can go from 0 to 2 and determines which variant is constructed.

Number of additional tanks	Variant	Notes
0	RS-1A	
1	RS-1B	
2	RS-1C	Stage requires boosters to perform properly

Guidance system

Guidance is provided by the four fins in the engine subassembly. Engine gimbaling is not used as the Reliant engine is fixed to its frame.

Recovery system

Stage recovery is provided by the parachutes. No control after stage separation is possible. Separation should take place when the propellant tanks are empty. Damage may occur if landing is attempted with propellant in the tanks.

Assembly instructions

If possible, use prefabricated subassemblies. While less precision is needed compared to earlier versions, prefab subassemblies ensure repeatable results. Also store a reference stage when complete.

Full bill of materials

- 4x AV-8R Winglet
- 1x LV-T30 "Reliant" Liquid Fuel Engine
- 2x FL-T400 Fuel Tank
- 2x Mk2-R Radial-Mount Parachute
- 1x TD-12 Decoupler

Extra FL-T400 Fuel Tank required for B and C variants.

Technology

This stage requires a level 4 technology level.

- Advanced Rocketry
 - Reliant engine
 - T400 tanks
- Flight Control
 - Fins
- Survivability
 - Parachutes
- Engineering 101
 - Decoupler

Fin subassembly

Part list

- 4x AV-8R Winglet
- 1x LV-T30 "Reliant" Liquid Fuel Engine
- 1x FL-T400 Fuel Tank

Assembly procedure

- Select LV-T30 engine as root part. Gas generator exhaust should face western orientation.
- Install FL-T400 axially above engine. Use dark "checkerboard" pattern.

- Install AV-8R winglets radially over tank using 4x symmetry and snap.
 - Orientation should be NE/SE/SW/NW.
 - Align fin height by matching second to lowest panel junction (line separating middle of fin) to exactly the centre of lower tank checkerboard line.
 - When aligned, lowest junction should point to end of engine shroud.

Parameters

Part count: 6

Subassembly cost: 4160 (full), 3976 (dry)

Subassembly mass: 3900 Kg (full), 1900 Kg (dry)



Figure 1: RS-1 fin subassembly. Red line highlights fin alignment to checkerboard pattern.

Parachute subassembly

Part list

- 1x FL-T400 Fuel Tank
- 2x Mk2-R Radial-Mount Parachute

Assembly procedure

- Select fuel tank as root.
 - Use dark "checkerboard" or white pattern.
- Install parachutes radially on 2x symmetry with snap.
 - Parachute orientation should be N/S.
 - Parachute tips should touch base of lower junction of rivet line panel on top of stage.

Parameters

Part count: 3

Subassembly cost: 1300 (full), 1116 (dry)

Subassembly mass: 2450 Kg (full), 450 Kg (dry)



Figure 2: RS-1 parachute subassembly. Red line goes from tip of parachute and shows overlap with "rivet panel" junction.

Final assembly

Part list

1x	RS-1 fin assembly
1x	RS-1 parachute assembly
1x	FL-T400 Fuel Tank (for B variant) or
2x	FL-T400 Fuel Tank (for C variant)
1x	TD-12 Decoupler

Assembly procedure

- Start with fin assembly
- Assemble extra tanks axially over fin assembly
- Connect parachute assembly axially over tanks
- Connect decoupler axially over upper parachute assembly tank

Stage configuration

Part	Variable	Value	Default	Notes
Fins	Pitch	Active	Yes	
	Yaw	Active	Yes	
	Roll	Active	Yes	
	Deploy	Retracted	Yes	
	Deploy direction	Normal	Yes	
	Deploy angle	0	No	
	Authority limiter	4	No	
Engine	Throttle	Main throttle	Yes	
	Thrust limiter	100	Yes	
LFOX tank 1	Flow priority	0	Yes	Uppermost tank
LFOX tank 2	Flow priority	+1	No	
LFOX tank 3	Flow priority	+2	No	If present
LFOX tank 4	Flow priority	+3	No	If present
Decoupler	Force percent	100	Yes	
	Enable crossfeed	False	Yes	
	Enable staging	True	Yes	
Parachutes	Deploy mode	When safe	Yes	
	Min pressure	0.6	No	
	Altitude	500	No	
	Spread angle	10	No	

Staging

Stage	Parts	Notes
0	Parachutes, decoupler	
1	Engine	

Parameters

Parameter	RS-1A	RS-1B	RS-1C
Mass (dry, kg)	2390	2640	2890
Mass (wet, kg)	6390	8640	10890
Cost (wet)	5660	6160	6660
Cost (dry)	5293	5609	5926
Part count	10	11	12

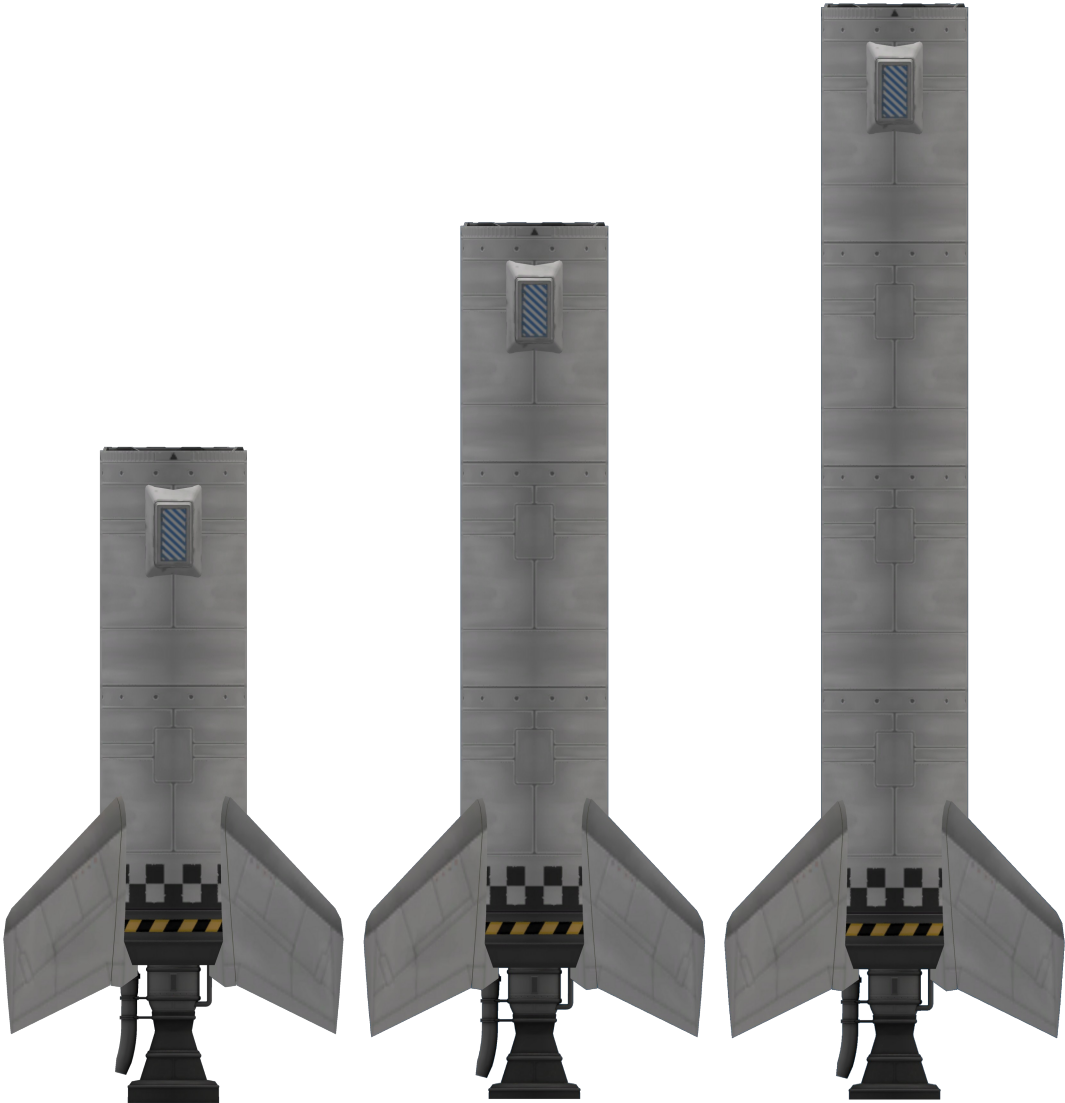


Figure 3: RS-1 family. RS-1A to the left, RS-1B at the center and RS-1C to the right.

Post flight maintenance

- RS-1 Decoupler
 - Enable staging
- Tanks
 - Fill

Pre flight checks

- Verify staging
- Verify parachute settings